

Omar Kamal

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RESEARCH STATEMENT

Multidisciplinary researcher with expertise in biomaterials, clinical research methodologies, and advanced software solutions for research applications. Passionate in combining materials and mechanical engineering principles with biological and medical applications to address complex healthcare and medicine challenges. Currently pursuing graduate research in electrospun biomaterials for drug-eluting surgical sutures, building on a background of designing controlled trials, conducting systematic reviews and meta-analyses, and developing custom software for research applications.

RELEVANT RESEARCH INTERESTS

Biomedical Engineering: Electrospinning, Nanofiber Yarn Fabrication, Drug-Eluting Sutures, Controlled Drug Delivery Systems, Epigenetics-Materials Interactions, Regenerative Medicine, Tissue Engineering, Medical Device Design, Biofabrication

Materials Science: Advanced Biomaterials, Biodegradable Polymers, Nanomaterials, Nanofabrication, Implant Materials, Piezoelectronics, Thermoelectronics, Semiconductors, Solar Cells

Research Methods: Biostatistics, Bioinformatics, Clinical Trial Design, Design of Experiments, Machine Learning for Biomedical Applications, Computational Methods, Software Development for Research Applications

EDUCATION

University of Alberta

Sep. 2025 – Present

Master of Science (Thesis) in Mechanical Engineering

Edmonton, AB

- **Thesis:** Drug-Eluting Electrospun PCL Coatings for Surgical Sutures
- **Relevant Coursework:** Machine Learning for Biomedical Applications, Statistical Mechanics with Applications, Design of Experiments in Mechanical Engineering, Colloidal Hydrodynamics, Nanofabrication, Nanocharacterization

University of Alberta

Sep. 2017 – Apr. 2023

Bachelor of Science in Materials Engineering

Edmonton, AB

- **Relevant Coursework:** Materials Characterization, Phase Transformations, Electrochemistry, Biomaterials, Bioengineering Design, Mechanical Properties, Materials Processing, Process Design, Failure Analysis, Thermodynamics, Cell Biology, Chemistry, Dielectric Materials, Semiconductors
- **Technical Skills:** Materials Selection, Computational Materials Science, Materials Modeling and Simulations, ThermoCalc, Porosity Measurement Methods, Processing Techniques, Data Analysis, Data Transformations

RESEARCH EXPERIENCE

University of Alberta

2025 – Present

Graduate Student, Department of Mechanical Engineering

Edmonton, AB

- **Drug-Eluting Electrospun PCL Coatings for Surgical Sutures (MSc Thesis):** Coating commercial monofilament surgical sutures (e.g., nylon) with a drug-loaded polycaprolactone (PCL) nanofiber sheath to reduce surgical site infections by releasing antibiotics (e.g., ciprofloxacin, tetracycline) or anti-inflammatories (e.g., ibuprofen) directly at the wound site.
- Adapting a dual-syringe yarn electrospinning setup with opposing-polarity needles and a rotating funnel to wrap nanofibers evenly around a non-conductive suture core, which conventional single-needle electrospinning coats poorly.

- Characterizing coating morphology and fiber diameter via scanning electron microscopy (SEM), and benchmarking tensile and knot-pull strength against USP suture requirements to confirm the coating preserves suture performance.
- Studying in vitro drug release kinetics across electrospinning parameter combinations, planned with design of experiments, to establish processing-structure-release relationships for scalable, cost-effective suture manufacturing.
- Previously designed a hollow-core yarn architecture using a water-soluble polyvinyl alcohol (PVA) sacrificial core as an alternative drug-loading route, which informed the current thesis scope.

Queen's University

2025 – Present

Research Assistant (Remote), Faculty of Family Medicine, under Dr. Roni Kraut

Kingston, ON

- **Clinical Analysis on Deprescribing:** Data collection and analysis using a standard checklist form implemented into a web service to analyze various implications for a broad range of medications.
- Collaborating remotely with the multidisciplinary research team to finalize manuscripts for peer-reviewed publication following Dr. Kraut's transition to Queen's University.

University of Alberta

2023 – 2025

Research Assistant, Department of Family Medicine, under Dr. Roni Kraut

Edmonton, AB

- **Pharmaceutical Interventions in Geriatric Care:** Contributed to published research on antihypertensive deprescribing protocols in long-term care settings, optimizing medication regimens for older adults. Applied statistical analysis techniques to evaluate efficacy and safety outcomes.
- **ADHD Assessment Optimization:** Designed and implemented controlled trials of various form designs to evaluate impact on ADHD diagnosis accuracy and efficiency in clinical settings. Created a digital platform with four distinct form versions to analyze how design influences diagnostic outcomes. Leveraging software development expertise to build custom research tools.
- **HPV Self-Sampling Implementation Study:** Contributed to a published implementation study evaluating the uptake and feasibility of HPV self-sampling among under-screened patients at a Canadian family medicine clinic.
- **Medical Device Efficacy Assessment:** Evaluated inhaler types, comparing dose delivery accuracy between inhalers with and without dose counters to improve patient medication management and reduce waste. Applied materials engineering principles to analyze device performance and reliability.
- Collaborated with multidisciplinary teams to develop research methodologies, analyze complex datasets, and prepare manuscripts for peer-reviewed publication. Integrated computational approaches to improve research efficiency and data quality.

PUBLICATIONS

Peer-Reviewed Journal Articles

- Wierenga, F., Dievert, H., Chiang, B., Yang, H., Babenko, O., Umali, M., Korownyk, C.S., Manca, D.P., Molstad, E., **Kamal, O.**, Kraut, R.Y. (2025). "Uptake and feasibility of HPV self-sampling among patients of a Canadian family medicine clinic." *Canadian Family Physician*, 71(11-12), e280-e289. DOI: [10.46747/cfp.711112e280](https://doi.org/10.46747/cfp.711112e280)
- Kraut, R.Y., Ono, C., Garrison, S., **Kamal, O.**, Doroshuk, M.L., Vandermeer, B., Amanna, G., Babenko, O. (2025). "Does the format of the adult ADHD self-report scale influence screen-positive rates? A randomized controlled trial in primary care." *Frontiers in Psychiatry*, 16, 1646293. DOI: [10.3389/fpsyt.2025.1646293](https://doi.org/10.3389/fpsyt.2025.1646293)

Published Conference Abstracts

- Kraut, R., Katz, M., Ning, S., Wilson, K., Durant, B., Attari, R., **Kamal, O.**, et al. (2025). "083 The need for inhalers to have dose counters: an observational study in Edmonton, Alberta, Canada." *BMJ Evidence-Based Medicine*, 30(Suppl 1), A42.
- Kraut, R., **Kamal, O.**, Vucenovic, A. (2024). "Deprescribing antihypertensives with weekly taper intervals in long-term care, secondary data analysis of OptimizeBP data." *The Annals of Family Medicine*, 22(Supplement 1), 6687. DOI: [10.1370/afm.22.s1.6687](https://doi.org/10.1370/afm.22.s1.6687)

Manuscripts in Preparation

- Kraut, R., et al. “Is breast density a prognostic factor of breast cancer mortality? A systematic review and meta-analysis of observational studies.” *In preparation*.

Conference Presentations & Posters

- **Kamal, O.**, Ayranci C., Tang T. (2026). “Hollow-Core PCL Yarn Fabrication for Biomedical Applications” Oral presentation at the Faculty of Engineering Graduate Research Symposium (FEGRS), University of Alberta, Edmonton, AB.
- Kraut, R. (2024). “Digital Form Design Impact on Clinical ADHD Assessment Efficiency.” Poster presented at the North American Primary Care Research Group, Quebec City, QC.

ENGINEERING PROJECTS

Binder Jetting & Sintering Process Optimization for QT 450-10 Ductile Iron 2023

Research Collaboration, Red Deer Polytechnic & University of Alberta *Edmonton, AB*

- Led materials characterization study using advanced porosity assessment techniques including gas/liquid pycnometry and image data processing to optimize additive manufacturing processes.
- Collaborated with academic researchers to implement and refine characterization methodologies for ductile iron, contributing to advancements in materials science.
- Analyzed process-structure-property relationships to optimize sintering parameters for enhanced mechanical properties.
- Developed custom software tools for image analysis and data processing, improving measurement precision and experimental throughput.

Process Design for Wear-Resistant Excavator Blade Teeth 2022

Capstone Project, University of Alberta *Edmonton, AB*

- Developed a comprehensive manufacturing process for high-performance wear-resistant components involving Ni-60 wt% Tungsten Carbide and 17-4 PH Stainless Steel.
- Performed detailed energy, material, and economic analyses to optimize the 50+ step manufacturing process, resulting in 15% cost reduction without compromising performance.
- Utilized ThermoCalc and other computational tools to predict and validate material properties and process parameters.
- Applied statistical design of experiments to efficiently identify optimal processing conditions, reducing development time by 30%.

RESEARCH SKILLS & METHODOLOGIES

Experimental Design: Clinical Trial Design, Comparative Studies, Randomized Controlled Trials, Blinded Studies, Crossover Studies, Design of Experiments

Data Analysis: Statistical Analysis, Meta-Analysis, Biostatistics, Data Visualization, Regression Analysis, Survival Analysis, Machine Learning for Biomedical Applications

Research Methods: Systematic Literature Review, Qualitative Research, Survey Design, Meta-analysis Research

Materials Characterization: Electrospinning, SEM/TEM, XRD, Tensile & Knot-Pull Testing, Porosity Assessment, Thermal Analysis, In Vitro Drug Release Kinetics

TECHNICAL SKILLS

Programming & Data Analysis: MATLAB, Python, Rust, Go, TypeScript/JavaScript, C, SQL, Statistical Analysis, Data Visualization, Image/Audio/Data Processing

Software & Tools: ThermoCalc, CAD Software, Materials Databases, Research Data Management Systems, Laboratory Information Management Systems

Programming for Research: Algorithm Development, Automated Data Collection, Digital Platform Creation, Research Tool Design, Statistical Software Implementation

Languages: Fluent in English; Proficient in Urdu/Hindi; Basic proficiency in German, Punjabi and Japanese

ADDITIONAL EXPERIENCE

Corsace

2015 – 2024

Founder, Lead Software Developer & Project Manager

Edmonton, AB

- Founded and led a fully volunteer-based, 200+ person international organization building large-scale online events.
- Developed data management systems and analytical tools for large-scale international events, handling complex datasets and implementing efficient data processing workflows.
- Created automation systems that reduced manual workload by 50% through implementation of database-driven solutions and process optimization.
- Led cross-functional teams and coordinated with stakeholders to ensure project milestones were met consistently and efficiently.
- Applied advanced statistical methods to analyze user engagement patterns and optimize platform performance.

Shifa Medical Clinic

2014 – 2017

Technical Assistant

Edmonton, AB

- Managed electronic medical record systems, ensuring 100% compliance with PIPEDA protocols while optimizing database performance for improved clinical workflow.
- Established secure protocols for patient data transfer between healthcare facilities, maintaining strict confidentiality standards and data integrity.
- Developed technical solutions for healthcare data management that reduced information retrieval time by 200% and improved clinical decision-making efficiency.